Interactive AI Driven Role Playing Game

Kaan Dai, Cem Ozan Doğan, Emir Şahin, Oğuzhan Şanlı

Advisor: Sajit Rao

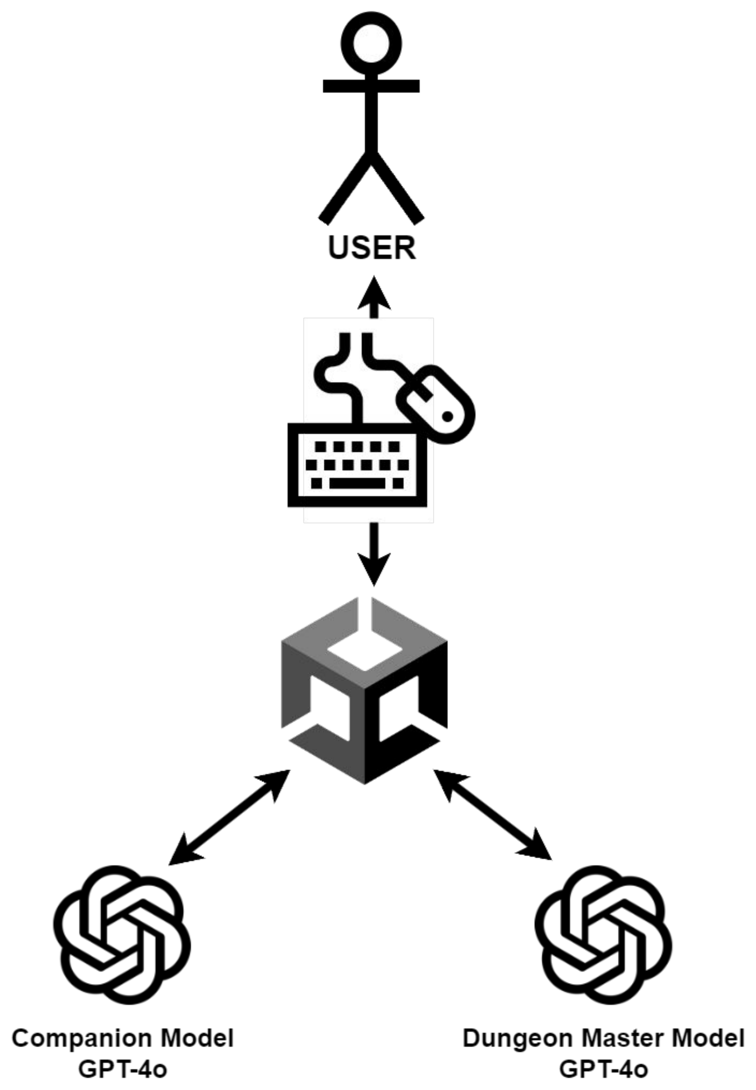
Department of Computer Engineering

Koç University

Project Description / Objectives

- The project aims to develop a unique, single-player, 2D role-playing game where players interact with AI companions in a dynamic world.
- The game uses player decisions and language models to create a narrative, with AI companions responding based on various factors like story, health, environment, tasks and relationship.
- The goal is to provide an immersive gameplay experience through real-time actions and text interactions.
- Our project aims to create a fun, engaging game with AI NPCs, targeting users seeking better game experiences.

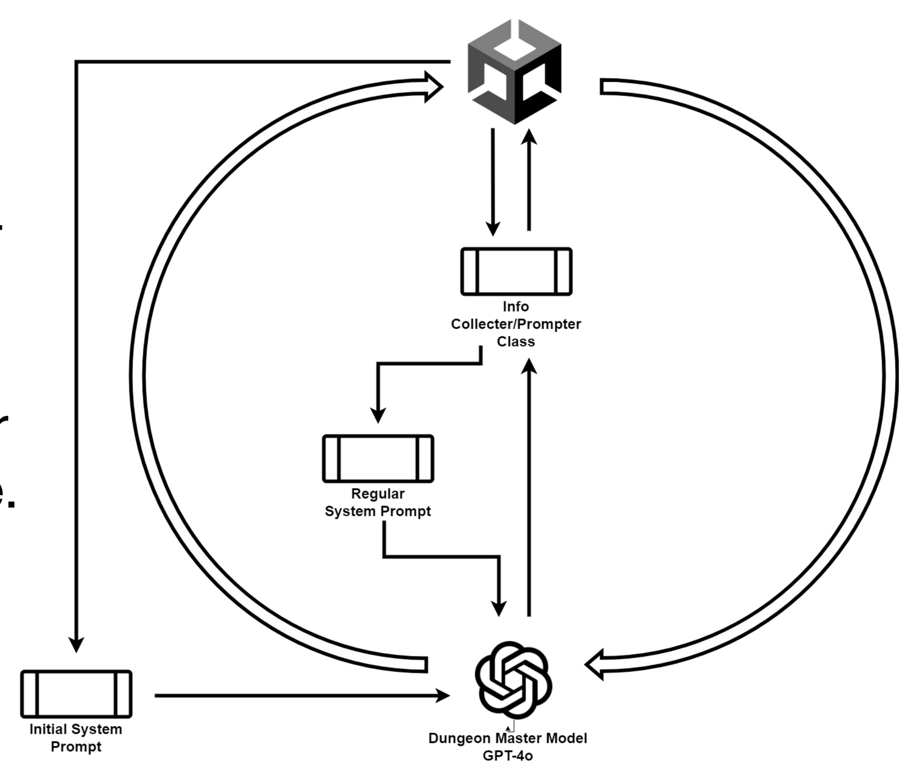
System Design

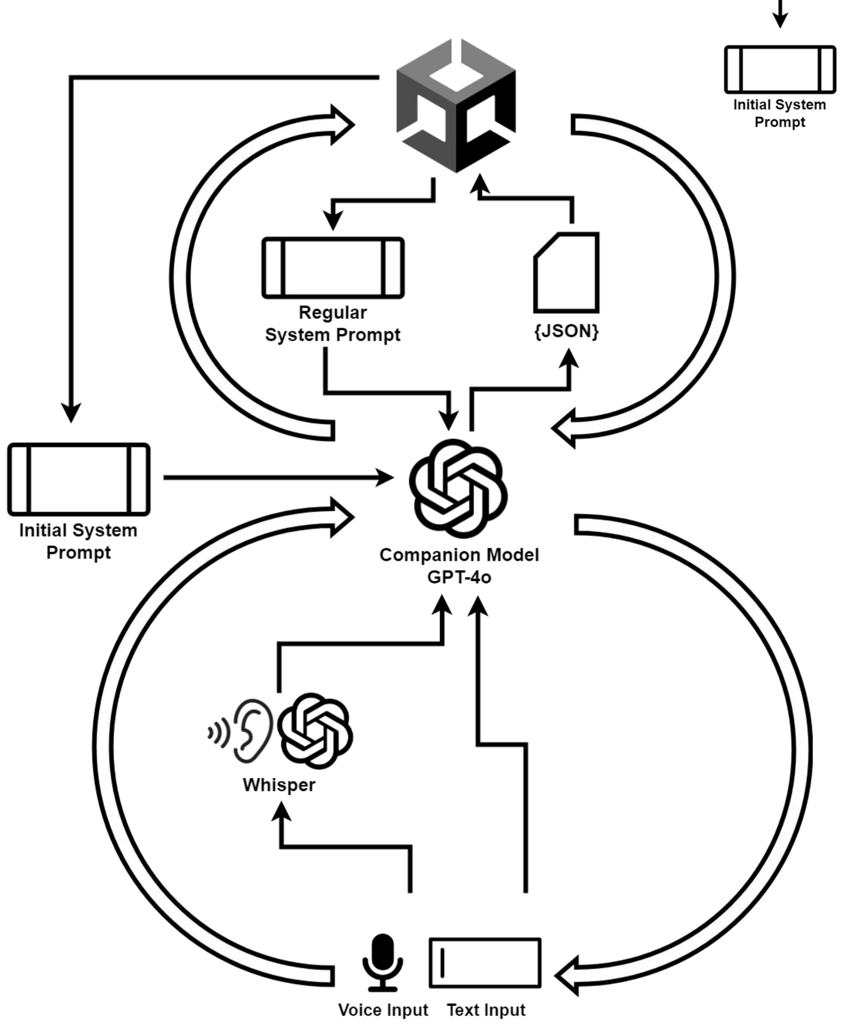


- In this diagram, we show a basic representation of our game system.
- User interacts with the Unity game engine via mouse and keyboard.
- Unity engine sends requests to the respective models.
- The results obtained from the models are transferred back to the user.

System Architecture

- In this diagram, Unity game engine calls the info collector class.
- Info collector class in turn prompts the Dungeon Master model to generate a narrative.





- On the left, there are two cycles.
- The top cycle visualizes the constant information exchange regarding the game environment between the companion model and the game engine
- The bottom cycle visualizes the continuous dialogue between the companion model and the player

Put your group photo (containing all members) here.

Put your QR Code here (right side of the photo).

Methods

- Game Engine: Unity
- Sprites: Unity Asset Store
- AI Models: OpenAI GPT-4o
- Voice Model: OpenAI Whisper
- Scripts: C#

Results

• We created a Platformer Role Playing Game, that utilizes 2 AI models: "The Dungeon Master Model" and "The Companion Model" within the Unity Game Engine.

FEATURES

The Player Can:

• Navigate the game environment with their keyboard and engage in combat with their mouse.

• Communicate with the companions by either using the chat button or with real-time voice detection.

• Read and hear the story created and narrated by the "Dungeon Master" model from the text at the top of the game screen.

• Access their inventory, see their items, their descriptions and can use these items whenever they want in the game to heal, increase their damage and armor statuses.

• Give directives, ask for items, command the companions to take action in the game.

• Build stronger relationship with the companions.

• Kill the enemy monsters and dungeon bosses.

• Collect the dropped items from the dungeon bosses when they are killed.

The difficulty of the levels are dynamically adjusted depending on how effectively the player completes them.

FUTURE WORK

- We would like to improve our project using more sophisticated methods to increase the AI driven story aspect of the game.
- Improving our quest design, taking into account story writing principles

Acknowledgement

We would like to thank Sajit Rao and Seher Özçelik for their guidance in this project.

References

[1] Sarge, "Prompt Engineering in ChatGPT for making NPCs!," *YouTube*, Jan. 23, 2023. https://www.youtube.com/watch?v=usrxIUGK9Gc (accessed May 16, 2024).